

Kulshrestha Chandhary

Kulshrestha
24/10/TC/013

JAWAHARLAL NEHRU UNIVERSITY
SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

Mid-Term Exam

M.CA. (Winter Semester-2024)

Course Name: Computer Architecture

Course Code: CS 106

Maximum Marks: 45

Total Time: 2 hour

Q1. a) Which among Harvard and Von Neumann Architecture is preferred and why?.

Three enhancements with following speedups are proposed for new architecture:

Speedup₁=30; Speedup₂=20; Speedup₃=15; only one enhancement is useable at time

(a) If enhancement 1 and 2 are each usable for 18% of the time, what fraction of the time must enhancement 3 be used to achieve overall speed up of 18?.

(b) Assume the enhancement can be used 30%, 40% and 30% of the time for enhancement 1, 2, 3 respectively. For what fraction of the reduced execution time no enhancement is used?.

(4+6)

Q2. (a) Which performance parameters can improve the execution time of the processor by improving the Computer Architecture?.

b) A weather forecasting computation requires 400 billion floating point operations. The problem is processed in a Super computer that can perform 80 megaflops. How long will it take to perform these calculations?

c) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.0 and ii) reduce CPI instruction type B to 2.1. Which one is worse?.

Type	CPI	Frequency
A	1.5	40%
B	2.5	40%
C	2.0	20%

(b
ca

(3+3+4)

- Q3. (a) Why there is a need for Memory Hierarchy in Computer Architecture. Explain with important parameters the Memory Hierarchy in Computer System?. How can one increase the relative size in a Memory system?.
- b) What is the difference between SRAM and DRAM cell. Explain with their working.

(5+5)

- Q4. (a) What are the total external connections in 256X16 Memory system?. What is the 90/10 rule?. What is the difference between Temporal and Spatial locality in memory system?.
- (b) Explain the difference between Direct, Full Associative and Set Associative cache mapping. Which one is preferred is preferred and why ?.

(5+3+2)

- Q5. (a) Consider a CPU with an average CPI as 1.8. Assume the instruction mix is ALU-35%, Load-15%, Store-20% Branch-30%. Assume the cache miss rate of 5% and miss penalty of 60 cycles ($=t_{MM}$). Calculate the effective CPI for a unified L1 cache.

(5)